Rohan Mahesh Rao

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SUMMARY

In the tech realm, my philosophy is simple: let the problem define the approach. I am passionate about full-stack development, AI, and machine learning, with a strong focus on building scalable, production-ready systems.

EXPERIENCE

JUNIOR SOFTWARE ENGINEER

KANGASYS PTY LTD **January 2024 - March 2026, Bangalore**

- Designed and delivered KangaKonnect, an in-house site onboarding platform, improving **product integration speed by 2×.**
- Led backend development for enterprise APIs using ASP.NET, owning end-to-end architecture from controllers to services, enforcing **SOLID** principles, and conducting code reviews.
- Architect and manage **AWS cloud infrastructure** for production systems using App Runner, ECS, ECR, and Lambda.
- Design and maintain **CI/CD pipelines** across dev, staging, and production environments for reliable deployments.
- Manage database performance for ClickHouse and PostgreSQL, including query optimization, indexing, and system stability for **billions of rows.**
- Drive **R&D initiatives** focused on AI-driven capabilities and advanced data processing.

SOFTWARE R&D INTERN

QUALITAS TECHNOLOGIES PVT LTD **October 2023 - January 2024, Bangalore**

- Developed and deployed **production-grade deep learning pipelines** using PyTorch and TensorFlow, integrating into existing CV systems.
- Implemented Vision Transformers (ViT) for classification and fault detection, **improving inference speed by 3×.**
- Built AI workflow automation systems using Semantic Kernel and OpenAI APIs, orchestrating **multi-step LLM pipelines** in Python.
- Contributed to cloud deployments on Microsoft Azure and supported full-stack integrations using the MERN stack.

PROJECT

ML-Assisted Image Despeckling

PES University • github.com/Rohanmrao/Image-Despeckling

- Built an ML-assisted despeckling pipeline for SAR images by training models (Random Forest, SVM, Multinomial & Bernoulli Naive Bayes) to predict optimal Lee filter kernel size from HSV features, improving noise reduction while preserving image details.
- Implemented post-processing with unsharp masking to enhance edge clarity, and evaluated performance on real SAR datasets to ensure practical effectiveness in noisy imaging conditions.

FRedBots

PES Innovation Lab • github.com/Rohanmrao/fredbots

- Developed a multi-agent reinforcement learning system for warehouse automation using Q-learning and ROS, enabling autonomous robots to perform real-time path planning, task allocation, and collision avoidance in a dynamic grid environment.
- Designed a scalable collaborative robotics framework with intelligent task scheduling and deadlock prevention via centralized controllers, improving efficiency of package delivery workflows in simulated warehouse operations (Gazebo + ROS).

EDUCATION

Bachelor of Technology, Electronics and Communication Engineering

Minor in Computer Science and Engineering • PES UNIVERSITY • 8.0 / 10.0

- 4× DAC (Distinction Award Certificate) holder – Semesters 1, 2, 5, 6.

INVOLVEMENT

Community Development Volunteer

Dept. of ECE, PESU

- Conducted computer literacy sessions and team-building activities for government school students.

PUBLICATIONS

FRedBots: Fundamental Reinforcement Learning Enabled Bots

IEEE • DOI: 10.1109/ICCCR65461.2025.11072598

SKILLS

ASP.NET, React, GenAI, MCP Servers.

MySQL, MongoDB, Elasticsearch, ClickHouse, PostgreSQL, Neo4j.

TypeScript, JavaScript, C#, Python, Shell Scripting.

Git, AWS, Docker, Kubernetes, Terraform, JIRA.